Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/14/2017
PI Name:	Olson, Sandra Ph.D.		
Project Title:	Correlation of Normal Gravity Aerospace Materials Flammability Data with Flammability in Microgravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Sandra.Olson@nasa.gov	**Fax:**	FY 216 977-7065
PI Organization Type:	NASA CENTER	**Phone:**	216-433-2859
Organization Name:	NASA Glenn Research Center		
PI Address 1:	LTX, Combustion Physics and Reacting Systems Branch		
PI Address 2:	MS 77-5, 21000 Brookpark Rd.		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2010	**End Date:**	12/31/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
Contact Email:	david.l.urban@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 12/31/2019, per PI (Ed., 1/26/2015) NOTE: Extended to 12/31/2014, per PI (Ed., 10/17/13) NOTE: Extended to 12/31/2013, per PI (Ed., 10/10/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hirsch, David (NTEC, Jacobs Engineering)		
Grant/Contract No.:	Internal Project		
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Performance Goal Text:			

Task Description:	NASA STD 6001 Test 1 is the major method used to evaluate flammability of materials intended for use in habitable environments of U.S. spacecraft. The method is an upward flame propagation test initiated in a stagnant environment and using a well-defined igniter flame at the bottom of a vertically mounted sample. A material passes this test if the vertical burn length is less than 15.2 cm (6 inches) and there is no evidence of transfer of burning debris. The test is conducted in the most severe flaming combustion environment (oxygen concentration, pressure) expected in the spacecraft, currently 30% oxygen at 10.2 psia. Recent research has shown that current normal gravity materials flammability tests do not correlate with flammability in ventilated, micro-, or reduced-gravity conditions. The materials selection for spacecraft is based on the assumption of commonality between ground test flammability results and spacecraft environments, which does not appear to be valid. To better understand the actual normal gravity flammability limits, a modified Test 1 protocol has been proposed that, as an alternative to qualifying materials as pass/fail in the worst-expected environments, measures the actual upward flammability limit (akin to the Limiting Oxygen Index (LOI)) for the material so that a more accurate assessment of the margin of safety of the material in the real use environment can be made. The method allows the option of selecting better or best materials based on the margin of safety, as opposed to what would be considered just "passing" from a flammability point of view. A research program is underway to correlate normal gravity flammability test data with data under ventilated reduced gravity or microgravity conditions. The program will determine materials suitability for microgravity and reduced gravity testing by evaluating oxygen concentration thresholds under various total pressures in normal gravity tests and in spacecraft-specific environments (ventilated microgravity or reduced partial gravity). The normal gravity data will be correlated with flammability data in microgravity environments and the safety factors involved when selecting spacecraft materials based on normal gravity testing will be evaluated. Rigorous correlations between ground test flammability data and data in ventilated spacecraft environments would allow selection of materials with increased fire resistance, and thus decrease the fire risk in space systems. The operational flexibility will also increase since the safety factors will be assessable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The identification of the fundamental limit in microgravity and its comparison with the normal gravity flammability limit will help us understand flammability in different configurations on Earth. It will improve the safety of astronauts in spaceflight so they can return safely to Earth.
Task Progress:	Normal gravity flame blowoff limits in an axisymmetric polymethylmethacrylate (pmma) rod geometry in upward axial stagnation flow were obtained in the laboratory apparatus. The blowoff limits are compared with microgravity Burning and Suppression of Solids – II (BASS-II) results recently obtained aboard the International Space Station. The lab testing used the same BASS-II concurrent rod geometry, but with the addition of normal gravity buoyant flow. Cast polymethylmethacrylate (pmma) rods of diameters ranging from 0.635 cm to 3.81 cm were burned in a vertical flow tube similar to an LOI device, but with the capability of varying the forced flow velocity. Oxygen concentrations ranging from 14 to 18% by volume were supplied from premixed bottles. The forced flow velocity where blowoff occurred was determined for each rod size and oxygen concentration. These normal gravity blowoff limits compare favorably with the BASS-II results when the buoyant stretch is accounted for and the flow is corrected by considering the blockage factor of the fuel. From these results, the normal gravity blowoff boundary for this axisymmetric rod geometry is determined as a function of oxygen concentration and equivalent stretch rate. The oxygen concentration at blowoff linearly increases as the equivalent stretch rate increases. This a new normal gravity ‘upward flame spread test’ method provides a flammability curve that can be extrapolated to the zero stretch limit to estimate the microgravity flammability limits—something current methods cannot do. This new test method can improve spacecraft fire safety for future exploration missions by providing a tractable way to obtain good estimates of material flammability in low gravity. Low-pressure blowoff experiments were also conducted with a stagnation flame stabilized on the forward tip of cast PMMA rods in a vertical wind tunnel. Pressure, forced flow velocity, gravity, and ambient oxygen concentration were varied. Stagnation flame blowoff is observed from a time-stamped video recording of the test. The blowoff pressure is determined from test section pressure transducer data that is synchronized with the video time stamp. The forced flow velocity is also determined from the choked flow orifice pressure. Most of the low pressure tests were performed in normal gravity, but a handful of microgravity drop tests were also conducted to determine the influence of buoyant flow velocity on the blowoff limits. The blowoff limits are found to have a linear dependence between the partial pressure of oxygen and the total pressure, regardless of forced flow velocity and gravity level. The flow velocity (forced and/or buoyant) affects the blowoff pressure through the critical Damkohler number residence time. For a constant critical Damkohler number, the partial pressure of oxygen at blowoff varies with the equivalent stretch rate (flow velocity) to provide this linear dependence. This is because the critical stretch rate increases linearly with increasing pressure at low pressure (sub-atmospheric pressures) since a second-order overall reaction rate with two-body reactions dominates in this pressure range.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Abstracts for Journals and Proceedings	Marcum JW, Olson SL, Ferkul PV. "BASS-II Microgravity Forced Flow Rod Blowoff Limits Compared to Normal Gravity Mixed Convective Blowoff Limits." 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. https://www.asgsr.org/ , Oct-2016
Abstracts for Journals and Proceedings	Marcum JW, Olson SL, Ferkul PV. "Correlation of normal gravity mixed convection blowoff limits with microgravity forced flow blowoff limits." 2016 Spring Technical Meeting Central States Section of the Combustion Institute, Knoxville, Tennessee, May 15-17, 2016. 2016 Spring Technical Meeting Central States Section of the Combustion Institute, Knoxville, Tennessee, May 15-17, 2016. http://cssci.org/meetings/2016knoxville/ , May-2016